

Relationship Between Tinnitus Loudness Measure by Visual Analogue Scale and Psychoacoustic Matching of Tinnitus Loudness

*Danuta Raj-Koziak, *Elżbieta Gos, *Weronika Świerniak, *Lucyna Karpiesz, *Iwona Niedziałek, *Elżbieta Włodarczyk, *Henryk Skarżyński, and *†‡Piotr H. Skarżyński

**World Hearing Center, Institute of Physiology and Pathology of Hearing, Warsaw/Kajetany; †Heart Failure and Cardiac Rehabilitation Department, Medical University of Warsaw, Warsaw; and ‡Institute of Sensory Organs, Warsaw/Kajetany, Poland*

Objective: The aim of the study was to assess the relationship between psychoacoustic matches of tinnitus loudness and tinnitus loudness measured with a visual analogue scale (VAS) in patients with normal hearing and patients with hearing loss.

Study Design: Cross-sectional study.

Patients: A clinical group of 140 adult patients (46.4% women, 53.6% men) aged from 19 to 81 years old who had had tinnitus for at least 6 months were included in the study. The most frequent reported localization of their tinnitus sensation was bilateral (48.6%); 40% experienced unilateral tinnitus; and 11.4% heard tinnitus in the head.

Methods: All participants were first asked to complete a VAS to indicate their tinnitus loudness. Hearing thresholds were then determined for each patient at frequencies from 0.125 to 8 kHz; loudness and frequency of the tinnitus were also matched psychoacoustically.

Results: Tinnitus loudness measured in dB SL was significantly lower in patients with bilateral hearing loss than in patients with unilateral hearing loss or in patients with normal hearing. Tinnitus loudness measured with VAS was significantly higher in patients with bilateral hearing loss than in patients with normal hearing. In patients with normal hearing there was a relationship between psychoacoustic matches of tinnitus loudness and tinnitus loudness measured with VAS, but this relationship did not hold for the hearing loss patients.

Conclusions: The VAS scale for tinnitus loudness does not generally correspond to psychoacoustic measures of tinnitus loudness. It is only indicative for tinnitus patients who have normal hearing. **Key Words:** hearing loss—tinnitus—visual analogue scale scale.

Otol Neurotol 40:16–21, 2019.

Subjective tinnitus is common, while objective tinnitus is rarely observed condition. Because tinnitus is mostly subjective, there are no clinical diagnostic tests to objectively measure the condition. If one is to diagnose tinnitus, there are two approaches: 1) focusing on the auditory characteristics of tinnitus, which means the perceptions of its pitch and loudness; 2) focusing on personal reactions to tinnitus, which means using a validated questionnaire or visual analogue scale (VAS) to gauge the negative impact that tinnitus has on a patient's functioning (1).

Psychoacoustic measures are able to define the auditory attributes of tinnitus using the psychoacoustic method of matching its pitch and loudness. Some think that characterizing tinnitus in this way is a poor indicator of tinnitus severity, and that it is impossible to gauge

tinnitus distress based on plotting the value on the audiogram (2–4). Nevertheless, although the clinical value of the psychoacoustic method of matching of tinnitus pitch and loudness is rather equivocal, it is still used as a counseling tool to assess tinnitus numerically (1).

Distress, handicap, and a patient's emotional reaction to tinnitus are often measured by using a validated tinnitus questionnaire, which is another way of quantifying the severity of tinnitus; more usefully, such a questionnaire—and there are several available—can assess changes in tinnitus severity over time, particularly after clinical intervention (5). The most frequently used questionnaires are the Tinnitus Handicap Inventory (THI) and the Tinnitus Functional Index (TFI) (6). The THI is a self-reported measure consisting of 25 items divided into three subscales (7,8). The TFI questionnaire is also a self-report measure used for evaluating tinnitus severity, and consists of 25 items divided into eight subscales (9).

A patient's own description rating of tinnitus loudness and annoyance—as measured on a VAS—is another possible approach to estimate either the patient's tinnitus

Address correspondence and reprint requests to Piotr H. Skarżyński, M.D., Ph.D., M.Sc., World Hearing Center, Institute of Physiology and Pathology of Hearing, Mokra 17 Street, Kajetany 05-830, Nadarzyn, Poland; E-mail: p.skarzynski@ifps.org.pl

The authors disclose no conflicts of interest.

DOI: 10.1097/MAO.0000000000002070

perception (loudness) or their negative reaction to the problem (annoyance) (10–12). VASs are a fast counseling tool, easy to understand and perform, and can be particularly useful at the stage of diagnosing tinnitus, especially at the first encounter. VASs have been used to assess changes in tinnitus loudness and annoyance due to treatment (13,14).

An important issue in measuring tinnitus perception is to check whether there is a correlation between measurements of tinnitus loudness based on VAS (TL-VAS) and tinnitus loudness matching psychoacoustically (TL-AM). At present, no work has been done which tries to match the degree of tinnitus severity corresponding to a given psychoacoustic measure of tinnitus loudness.

AIM

The aim of this study was to assess the relationship between TL-AM and a TL-VAS. We investigated this relationship in three patient groups: those with normal hearing and those with either unilateral or bilateral hearing loss.

MATERIAL

In total, 140 adult patients (46.4% women, 53.6% men) underwent evaluation of their chronic tinnitus at the Audiology and Phoniatrics Clinic of the Institute of Physiology and Pathology of Hearing, Warsaw. The participants ranged in age from 19 to 81 years (mean = 50.01; standard deviation [SD] = 14.24). All reported that their tinnitus sensation had lasted for at least 6 months, with a range of 0.5 to 30 years (mean = 5.55; SD = 5.94). In terms of localization, the most frequently reported site was bilateral (48.6%); 28.6% reported unilateral tinnitus in the left ear; 11.4% in the right; and 11.4% heard tinnitus in the head. Characteristics of the patients are shown in Table 1.

TABLE 1. *Characteristics of tinnitus patients*

Study Group (N)	140
Age (yr)	<i>M</i> = 50.01 (SD = 14.24)
Gender	
Male	75 (53.6%)
Female	65 (46.4%)
Localization of tinnitus	
Both ears	68 (48.6%)
Left ear	40 (28.6%)
Right ear	16 (11.4%)
Head	16 (11.4%)
Duration of tinnitus (yr)	5.55 (SD = 5.94)
Onset of tinnitus	
Gradual	75 (53.6%)
Sudden	60 (42.9%)
Not known	5 (3.5%)

M indicates mean; SD, standard deviation.

METHOD

Hearing thresholds were determined for the right and left ear of each patient at frequencies of 0.125, 0.25, 0.5, 1, 2, 4, and 8 kHz. Normal hearing, according to the Bureau International d'Audiophonologie (BIAP) recommendation (15) is defined as an air-threshold value of 20 dB HL or less at all tested frequencies. Unilateral hearing loss was recognized when there was normal hearing in one ear and hearing loss in the other with threshold values greater than 20 dB at one or more frequencies. Thresholds in both ears greater than 20 dB at one or more frequencies were defined as bilateral hearing loss.

All participants first completed a Visual Analogue Scale measurement of tinnitus loudness (VAS-L). The purpose and use of the VAS-L was explained to the patient, who answered the question: "How loud was your tinnitus during the last week?". The endpoints for the VAS-L scale were designated as "inaudible" at the left end of the scale and "extremely loud" at the right. Intermediate feelings were marked somewhere between endpoints, depending on loudness. During analysis, results were scaled from 0 to 100. Psychometric properties of patient-reported VASs for measuring subjectively perceived tinnitus loudness, annoyance, distress, and degree of coping have already been evaluated in a Polish setting (12).

The psychoacoustic loudness and frequency of each patient's tinnitus was evaluated by presenting sounds designed to be similar to those described by the patient. Pure-tone or narrow-band noise was presented over headphones at each frequency from 0.125 to 12.5 kHz from an audiometer 10 dB SL above the participant's hearing threshold. As soon as the individual was able to identify the sound as being most similar to their tinnitus, he or she raised their hand. When a tinnitus pitch match was made, the signal was increased from the hearing threshold level by 5 dB SL, or less if required, and at each step the individual was asked if the loudness of the sound matched their tinnitus. The measurement of tinnitus loudness was performed twice with a short break between the first and second measurement. The lower intensity measure was used in the study. The sounds were presented to the ear contralateral to the site of the tinnitus in patients with unilateral tinnitus. In patients with bilaterally asymmetrical tinnitus the sounds were presented to the side where tinnitus was subjectively less loud. If the tinnitus was bilaterally symmetrical or experienced in the head, the patient themselves selected the ear to be tested.

The study has been approved by the Ethical Committee of the Institute of Physiology and Pathology of Hearing (IFPS:KB/10/2015) and conforms to the stipulations of the Declaration of Helsinki.

STATISTICAL ANALYSIS

A Kruskal–Wallis test was used to test differences in TL-AM and TL-VAS between the three groups of patients (normal hearing, unilateral hearing loss, and bilateral hearing loss). This analysis was followed with a pairwise comparison (the Mann–Whitney *U* test) with Bonferroni correction (0.05 divided by three because of the number of tests conducted).

The relationship between TL-AM and TL-VAS was calculated using Pearson's *r* correlation coefficient. Criteria provided by Cohen (16) were used to interpret the strength of correlation: very strong (0.5–1.0), moderate (0.3–0.5), or weak (<0.3). Linear regression analysis was conducted to test whether tinnitus loudness

measure by VAS could be predicted from matched tinnitus loudness (16).

RESULTS

Hearing Status and Tinnitus Localization

Based on the results of the pure tone audiometry thresholds, patients were assigned to one of three categories: normal hearing (33.6%), hearing impaired with unilateral hearing loss (22.9%), and bilateral hearing loss (43.5%).

For subjects with normal hearing, most (51.1%) localized their tinnitus in one ear, 31.9% localized it in both ears, and 17% in the head. For patients with unilateral hearing loss, tinnitus was reported as coming from both ears (46.8%), one ear (43.8%), or in the head (9.4%). Patients with bilateral hearing loss reported tinnitus in both ears (62.3%), in one ear (29.5%), or in the head (8.2%).

Tinnitus Loudness Measure by Psychoacoustic Matching (TL-AM) in Relation to Hearing Status

Matched tinnitus loudness in the study group ranged from 0 to 68 dB SL with an average $M=18.13$ (SD=15.69). The average level of tinnitus loudness calculated for the three groups (normal hearing, unilateral hearing loss, and bilateral hearing loss) is shown in Figure 1A.

The tinnitus loudness in dB SL for the group of patients with normal hearing was $M=24.38$ dB SL (SD=18.92); for patients with unilateral hearing loss $M=21.06$ dB SL (SD=14.46); and for patients with bilateral hearing loss $M=11.77$ dB SL (SD=10.49). The difference between the groups was statistically significant ($H=17.29$; $p<0.001$).

Pairwise comparisons with adjusted p -value showed that patients with bilateral hearing loss reported lower tinnitus loudness than patients with normal hearing ($U=865.0$; $p<0.001$) and lower tinnitus loudness than

patients with unilateral hearing loss ($U=559.5$; $p<0.001$). There was no significant difference between subjects with normal hearing and subjects with unilateral hearing loss.

Tinnitus Loudness Measure by VAS (TL-VAS) in Relation to Hearing Status

Subjectively perceived tinnitus loudness in the study group as measured with VAS-L ranged from 0 to 100 points with the average $M=55.19$ (SD=26.09).

The level of tinnitus loudness was compared between the three groups of the patients. Its average level was calculated and shown in Figure 1B.

The loudness of tinnitus for the group of normal hearing patients was $M=46.43$ (SD=27.38); for patients with unilateral hearing loss $M=59.66$ (SD=27.15); and for patients with bilateral hearing loss $M=59.59$ (SD=23.05). The difference between the groups was statistically significant ($H=8.11$; $p=0.017$). Patients with bilateral hearing loss reported higher tinnitus loudness than patients with normal hearing ($U=1010.50$; $p=0.009$). Other differences were not statistically significant.

Correlation Between Tinnitus Loudness Measure by Psychoacoustic Matching (TL-AM) and Tinnitus Loudness Measure by VAS (TL-VAS)

Tinnitus loudness (TL-AM) did not meet the assumption of a normal distribution and was therefore converted into a logarithm for correlation and regression analyses. Correlation between TL-AM and TL-VAS in all the patients was statistically significant but weak ($r=0.17$; $p=0.039$). There was no significant relationship in the group of patients with unilateral hearing loss ($r=0.06$; $p=0.733$) or in the patients with bilateral hearing loss ($r=0.14$; $p=0.270$). However, in normal hearing patients the correlation was statistically significant ($r=0.46$; $p=0.001$). The relationship was positive, i.e., the higher the tinnitus loudness measure by

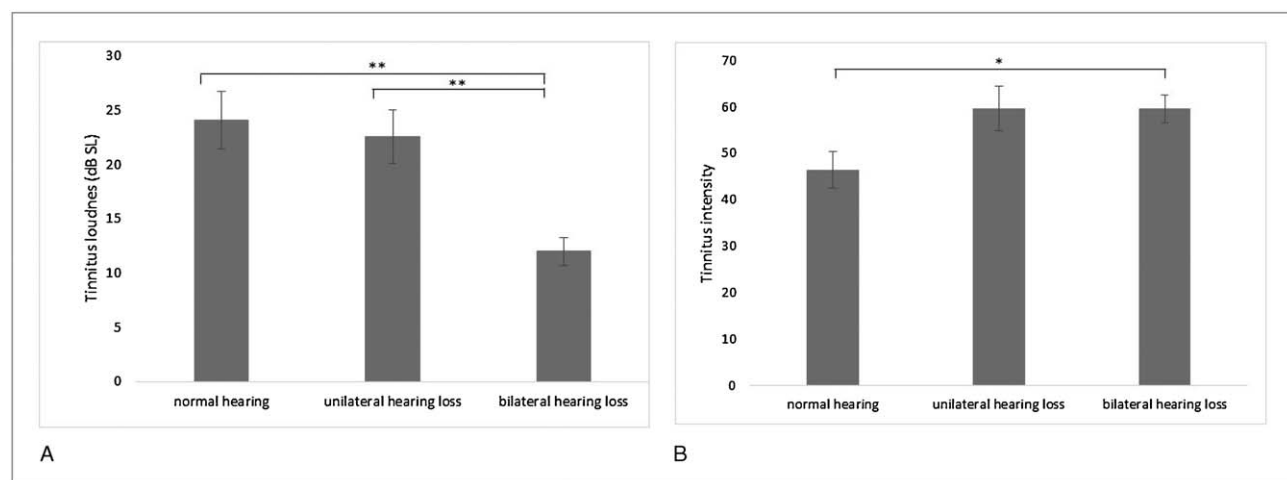


FIG. 1. Tinnitus loudness (in dB SL) (A) and visual analogue scale results of tinnitus loudness (B) in patients with normal hearing and patients with unilateral and bilateral hearing loss. The bars represent mean scores; the error bars represent standard error; ** $p<0.001$.

psychoacoustic matching (TL-AM), the higher tinnitus loudness measure by VAS (TL-VAS).

Tinnitus Loudness Measure by Psychoacoustic Matching (TL-AM) as a Predictor of Tinnitus Loudness Measure by VAS (TL-VAS)

Linear regression analysis was conducted to investigate whether subjectively perceived tinnitus loudness measured by VAS could be predicted by tinnitus loudness measured by psychoacoustic matching. TL-VAS was the dependent variable and TL-AM was treated as a potential predictor. The analysis was performed in the group of all patients and separately in the three groups with different hearing status. The results are shown in Table 2.

The regression model was statistically significant in the group of all patients, but only 3% of the variability of TL-VAS was explained by TL-AM.

The regression model was statistically significant in the group of subjects with normal hearing where nearly 23% of the variability in TL-VAS was explained by TL-AM. Standardized regression coefficient was significant only in this group ($\beta = 0.46$; $p = 0.001$). A high level of tinnitus loudness measure by VAS could be predicted from a high level of tinnitus loudness measure by psychoacoustic matching, but only in the case of subjects with normal hearing.

In the case of the subjects with hearing loss (unilateral or bilateral), TL VAS could not be predicted by TL-AM.

DISCUSSION

The results of matching tinnitus loudness to an external sound (in dB SL) indicated that patients with unilateral and bilateral hearing loss reported lower tinnitus loudness than those with normal hearing. For individuals with bilateral hearing loss the difference in loudness compared with a group of individuals with normal hearing may be explained by recruitment. Loudness recruitment is a common finding in patients with cochlear hearing loss caused by outer hair cell dysfunction, and is characterized by a large increase in perceived loudness caused by a slight increase in a sound's intensity. Recruitment does not depend on mood or level of anxiety (17). The mechanism underlying volume compensation is unclear, but it is thought that it may be due to nonlinearity of electro-mechanical transduction in outer hair cells (18–20).

If the group of patients with tinnitus and hearing loss had loudness recruitment, they would report tinnitus loudness as much quieter than those with normal hearing. Results of tinnitus loudness matching were affected in this group by cochlear hearing loss, which is accompanied by loudness recruitment, whereas in the group with normal hearing tinnitus loudness rating reflected the actual intensity of the tinnitus. In comparison, the subjectively perceived tinnitus loudness in patients with hearing loss, as measured with VAS, was significantly higher than the tinnitus loudness as measured by matching (dB SL). The inference is that the results of tinnitus loudness measure by psychoacoustic matching in patients with hearing loss are underestimated because of loudness recruitment. Similar results have been reported in patients where tinnitus loudness was matched at the frequency of tinnitus (21,22). In these reports, sensation levels were generally lower at the frequency of the tinnitus than at frequencies where hearing was normal.

Psychoacoustically, it has been found that tinnitus loudness typically matches just a few dB above the threshold of hearing, whereas the subjectively perceived loudness of tinnitus is often reported to be intolerable (23,24). Our investigations confirm these results, in that the loudness psychocoustically matching was significantly lower than the tinnitus loudness as rated with the VAS scale—but only in patients with tinnitus and hearing loss.

In general, loudness matching has no clinical value for diagnosing tinnitus or measuring the results of tinnitus treatment. However, it has been reported that tinnitus loudness matching can be used as a tool for measuring changes in tinnitus loudness before and after surgery for vestibular Schwannoma (25). Loudness matching has been used also by clinicians to show patients the relation of tinnitus loudness to the threshold of hearing, an approach intended to convince the patient that tinnitus is not so loud as it seems (1).

The VAS method of measuring tinnitus loudness indicates both the perception of tinnitus and the patient's reaction to it. Psychometric evaluation of VAS in a Polish setting has shown that there are moderate to strong correlations between results of the Tinnitus Handicap Inventory (THI), the Tinnitus Functional Index (TFI), and subscale A of the Tinnitus and Hearing Survey Questionnaire (THS, subscale A), confirming the construct validity of VAS (26–29). VAS measurements have also been reported to be significantly correlated with

TABLE 2. Results of regression analysis for tinnitus loudness measure by VAS (dependent variable) with tinnitus loudness measure by psychoacoustic matching as a predictor

	<i>R</i>	<i>R</i> ²	SE	<i>F</i>	<i>p</i>
All patients	0.174	0.030	25.79	4.32	0.039
Normal hearing	0.458	0.210	24.60	11.97	0.001
Unilateral hearing loss	0.063	0.004	27.54	0.01	0.733
Bilateral hearing loss	0.143	0.021	23.00	0.52	0.270

F indicates value of *F*-test; *p*, *p*-value; *R*, Pearson's correlation; *R*², coefficient of determination; SE, standard error of the estimate.

tools measuring tinnitus-related distress (11). The VAS approach has been used not only as a fast counseling tool, but also to assess changes in tinnitus loudness and annoyance due to treatment, such as with gabapentin (13).

The results of our study indicate that patients with bilateral hearing loss perceive tinnitus loudness (as measured by VAS) at higher loudness levels than in patients with normal hearing. How can this result be explained? Adamchic et al. (10) expressed the view that psychoacoustic matching procedures rely mainly only on sensory judgement of tinnitus loudness, whereas the rating of subjectively perceived tinnitus loudness also depends on emotional and cognitive factors. It has also been confirmed that a patient's response on a visual scale will correlate with the degree to which the person is impacted by tinnitus (30). So higher VAS ratings in patients with hearing loss compared with normal hearing patients could reflect problems with both tinnitus and hearing loss. We suggest that VAS measures of tinnitus loudness in patients with hearing loss might be over-reported because of overlapping complaints related to tinnitus and hearing loss. Ratnayake et al. (31) described how hearing impairment can contribute significantly to the distress caused by tinnitus. Therefore, in terms of a questionnaire, hearing impairment may make the tinnitus problem appear worse than it really is (1).

LIMITATIONS OF THE STUDY

A limitation of this work arose from our decision to use only VAS ratings of tinnitus loudness and not annoyance. A more complete picture would come from relating matched tinnitus loudness to tinnitus annoyance as measured on a VAS scale.

CONCLUSIONS

By assessing correlations between psychoacoustic matching of tinnitus loudness and the results of Visual Analogue Scale measurement of tinnitus loudness, in patients with normal hearing, unilateral hearing loss, and bilateral hearing loss, it is concluded that the evaluation of tinnitus loudness with VAS could be a useful tool in diagnosing patients and measuring effects of treatment—but only in normal hearing patients.

The VAS scale for tinnitus loudness does not generally correspond to psychoacoustic measures of tinnitus loudness. It is only indicative for tinnitus patients with normal hearing.

REFERENCES

1. Henry JA. Measurement of Tinnitus. *Otol Neurotol* 2016;37: 276–285.
2. Zeman F, Koller M, Schecklmann M, et al. Tinnitus assessment by means of standardized self-report questionnaires: psychometric properties of the Tinnitus Questionnaire (TQ), the Tinnitus Handicap Inventory (THI), and their short versions in an international and multi-lingual sample. *Health Qual Life Outcomes* 2012;10:128.
3. Andersson G, Porsaeus D, Wiklund M, et al. Treatment of tinnitus in the elderly: a controlled trial of cognitive behavior therapy. *Int J Audiol* 2005;44:671–5.
4. Jakes SC, Hallam RS, McKenna L, et al. Group cognitive therapy for medical patients: an application to tinnitus. *Cogn Ther Res* 1992;16:67–82.
5. Meikle MB, Stewart BJ, Griest SE, et al. Assessment of tinnitus: measurement of treatment outcomes. *Prog Brain Res* 2007;166: 511–521.
6. Baguley D, McFerran D, Hall D. Tinnitus. *Lancet Lond Engl* 2013;382:1600–7.
7. Newman CW, Jacobson GP, Spitzer JB. Development of the tinnitus handicap inventory. *Arch Otolaryngol Head Neck Surg* 1996;122:143–8.
8. Newman CW, Sandridge SA, Jacobson GP. Psychometric adequacy of the Tinnitus Handicap Inventory (THI) for evaluating treatment outcome. *J Am Acad Audiol* 1998;9:153–60.
9. Meikle MB, Henry JA, Griest SE, et al. The tinnitus functional index: development of a new clinical measure for chronic, intrusive tinnitus. *Ear Hear* 2012;33:153–76.
10. Adamchic I, Langguth B, Hauptmann C, et al. Psychometric evaluation of visual analog scale for the assessment of chronic tinnitus. *Am J Audiol* 2012;21:215–25.
11. Figueiredo RR, Azevedo AA, Oliveira Pde M. Correlation analysis of the visual-analogue scale and the Tinnitus Handicap Inventory in tinnitus patients. *Braz J Otorhinolaryngol* 2009;75:76–9.
12. Raj Kozia D, Gos E, Świerniak W, et al. Visual Analog Scales as a tool for assessment of the tinnitus: psychometric evaluation in a clinical population, in press.
13. Aazh H, El Refaie A, Humphriss R. Gabapentin for tinnitus: a systematic review. *Am J Audiol* 2011;20:151–8.
14. Vanneste S, Plazier M, van der Loo E, et al. The neural correlates of tinnitus-related distress. *NeuroImage* 2010;52:470–80.
15. BIAP - Bureau International d'Audiophonologie. Available at: <http://www.biap.org/recom02-1.htm>. Accessed January 8, 2018.
16. Cohen J. *Statistical Power Analysis for the Behavioral Sciences*. New York: Routledge Academic; 1988. 79–80.
17. Fowler EP. The use of threshold and louder sounds in clinical diagnosis and the prescribing of hearing aids. New methods for accurately determining the threshold for bone conduction and for measuring tinnitus and its effects on obstructive and neural deafness. *Transact Am Otol Soc* 1938;28:154–71.
18. Preyer S, Bootz F. Tinnitus models for use in tinnitus counselling therapy of chronic tinnitus patients. *HNO* 1995;43:338–51.
19. Zhang J, Zhang Y, Zhang X. Auditory cortex electrical stimulation suppresses tinnitus in rats. *J Assoc Res Otolaryngol* 2011;12: 185–201.
20. Tonndorf J. Stereociliary dysfunctions, a cause of sensory hearing loss, recruitment, poor speech discrimination and tinnitus. *Acta Otolaryngol (Stockh)* 1981;91:469–80.
21. Goodwin PE, Johnson RM. The loudness of tinnitus. *Acta Otolaryngol* 1980;90:353–9.
22. Henry JA, Flick CL, Gilbert A, et al. Reliability of tinnitus loudness matches under procedural variation. *J Am Acad Audiol* 1999; 10:502–20.
23. Meikle MB, Vernon J, Johnson RM. The perceived severity of tinnitus. Some observations concerning a large population of tinnitus clinic patients. *Otolaryngol Head Neck Surg* 1984;92: 689–696.
24. Zenner HP, Pfister M, Birbaumer N. Tinnitus sensitization: sensory and psychophysiological aspects of a new pathway of acquired centralization of chronic tinnitus. *Otol Neurotol* 2006;27:1054–63.
25. Cope TE, Baguley DM, Moore BC. Tinnitus loudness in quiet and noise after resection of vestibular schwannoma. *Otol Neurotol* 2011;32:488–96.
26. Skarzynski PH, Raj-Kozia D, Rajchel JJ, et al. Adaptation of the Tinnitus Handicap Inventory into Polish and its testing on a clinical population of tinnitus sufferers. *Int J Audiol* 2017;56:711–5.
27. Wrzosek M, Szymiec E, Klemens W, et al. Polish translation and validation of the Tinnitus Handicap Inventory and the Tinnitus Functional Index. *Front Psychol* 2016;7:1–11.

28. Raj-Koziak D, Gos E, Rajchel JJ, et al. Tinnitus and Hearing Survey: a Polish study of validity and reliability in a clinical population. *Audiol Neurotol* 2017;22:197–204.
29. Henry JA, Griest S, Zaugg TL, et al. Tinnitus and hearing survey: a screening tool to differentiate bothersome tinnitus from hearing difficulties. *Am J Audiol* 2015;24:66–77.
30. Zenner HP, De Maddalena H. Validity and reliability study of three tinnitus self-assessment scales: loudness, annoyance and change. *Acta Otolaryngol* 2005;125:1184–8.
31. Ratnayake SAB, Jayarajan V, Bartlett J. Could an underlying hearing loss be a significant factor in the handicap caused by tinnitus? *Noise Health* 2009;11:156–60.